

ANIVAE-2025: Workshop Series on Animation in Virtual and Augmented Environments

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Figure 1: Images taken from selected ANIVAE-2025 workshop papers.

ABSTRACT

As multifaceted medium, animation has expanded beyond its traditional associations with cinema and television. In emerging fields like augmented and virtual reality, animation has become integral to artistic, scientific, and commercial applications. The ANIVAE workshop series brings together experts from animation, media studies, and computer science to explore how animation will transform immersive experiences. This interdisciplinary forum fosters the development of new paradigms and techniques for integrating animation into AR/VR. By showcasing innovative research and applications, ANIVAE is advancing both artistic expressions and technological innovations in immersive media. This front matter provides an overview of the latest contributions to the field, highlighting how animation in AR/VR is pushing the boundaries of art-based research, augmented exhibitions, and motion-capture.

Index Terms: Animation, Virtual Reality, Augmented reality, Art-based Research, Media art, Stop-motion.

1 INTRODUCTION & SCOPE

Animation is a comprehensive and heterogeneous form of media. Often associated with cinema and television, animated images have entered many areas of life, particularly games, installations, or data visualizations, taking on operative, communicative, epistemic, and didactic tasks, among other things. In this context, Suzanne Buchan [1] speaks of pervasive animation, a media world in which animated images are omnipresent. In one of the most innovative fields of media production, augmented (AR) and virtual reality (VR), animation is an integral part of artistic, scientific, and economic applications in animated forms as well as mixed with live-recorded footage. In both fields, generative AI is applied in various aspects of production. Within the animation process, it assists, for example, in

story development and animatics, character design and animation, or virtual-scene creation. For AR/VR it is not only useful for procedural content creation, but also enhances interactive and dynamic environments, scene and context understanding, and creates more complex avatars or non-player characters.

Connecting specialists from various digital humanities research areas (such as animation, games, and media studies) with experts from vision-oriented computer science areas (such as computer graphics, information visualization or AI) and experts from technically oriented computer science areas (such as data integration, internet of things, or smart automation), the ANIVAE workshop series aims to create an open and exciting environment. By encouraging synergies of interdisciplinary approaches, the workshop maps animation within the AR/VR context from different angles and creates new knowledge in this research field.

2 CURRENT CHALLENGES

ANIVAE wants to advance the state-of-the-art research in digital humanities with (software) design and visualization for AR/VR systems, to shape a common understanding, to compare systems and approaches, and derive common paradigms, to develop useful and necessary methods and techniques, and to foster new ideas.

This interdisciplinary approach fosters a deeper understanding of how animation techniques can be integrated into AR/VR environments, enhancing immersion and storytelling. Motion and performance capture become vital tools for creating lifelike interactions and emotional resonance within these spaces. The tactility and physicality of AR/VR add new dimensions to digital materiality, allowing for a more visceral engagement with animated content.

In these environments, animated content serves a variety of use cases, from interactive media art to narrative-driven AR/VR films, demonstrating how these technologies reshape storytelling. The interplay of hardware and software becomes crucial for supporting animation workflows in AR/VR, while AI-driven processes can optimize real-time interaction and content generation. By combining media art, animation, and cutting-edge AR/VR technologies, new interdisciplinary and intermedia approaches emerge, advancing both artistic expression and technological innovation in the realm of immersive experiences.

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3 SUMMARY OF ACCEPTED PAPERS

This section briefly summarizes the contributions of the papers accepted for the 2025 edition of the ANIVAE workshop series.

3.1 Animated Augmented Realities

Two contributions address the topic of animation and augmented reality. The paper *Augmenting Murals: Creating Playful AR Experiences for Wall Art Exhibitions* explores the use of AR to enhance outdoor art exhibitions in natural settings, focusing on overcoming the limitations of traditional marker-based AR systems. It introduces a markerless AR approach utilizing geospatial tracking, which enables more flexible and user-friendly interactions in large-scale outdoor settings. A mobile AR application is presented, designed to augment murals with animations and interactive features, showcasing innovative ways to create engaging, user-centered art experiences. The paper *Animating Nature Beyond the Screen: Representation of Nature in AR Animation* examines the use of AR and animation to address nature-related themes. By merging AR and animation, the study highlights their enhanced potential across fields such as art, commerce, and environmental advocacy. It reviews the evolution of nature representation in animated films, from scientific time-lapse techniques to the integration of immersive technologies. The paper also investigates the *Garden of the Anthropocene* [2] by Tamiko Thiel, an ecological AR installation that exemplifies the potential of these technologies to make environmental issues more accessible and engaging.

3.2 Animated Art-based Research

The following two contributions combine stop-motion animation techniques with VR to explore the intersection of art, technology, and perception through a series of experimental workshops.

The work of *Fragmenting Space and Time: Bridging Kinetic Sculptures and Pre-Cinematic Concepts for Virtual Reality* is inspired by an expanded stop-motion concept and by Marey's chronophotographic gun and Edison's kinoscope. In collaboration with the artist Anna Vasof two VR environments were created. For *Shooting Marey* the VR environment is a minimalist black space with a hemispherical grid resembling a birdcage. Users control a virtual camera to find and capture birds, represented as frames from Marey's motion studies. Each photo triggers an external capture of the participant's movements. The final sequence shows two perspectives: the inside view with animated bird images and the outside view with the user's interaction. In the rotating kinetic sculpture *The Cage of Time* users are immersed in an illusion of time through space fragmentation and stop-motion animation, with the ability to break the illusion by closing one eye. Both artworks raise key questions about how pre-cinematic techniques can be extended to VR, how tangible objects and immersive experiences can be merged, and how unconventional interfaces can create novel VR interaction modalities.

The paper *Artistic Exploration of a Pixilated Choreography in Virtual Reality: Exploring 3D Gaussian Splatting for Stop-Motion Animation* analyzes a novel pixilation technique for VR, using 35 synchronized smartphones to capture approximately three minutes of animation over five days. This setup allowed for varied scales, with miniature backgrounds and full-size actors. Gaussian splatting was employed for an artistic, non-photorealistic aesthetic, enhancing the fragmented movement of pixilation and offering potential for further creative manipulation. The dramaturgical structure of the project developed by animation artist Paul Wenninger includes a loop between two scenes with the performer Raul Maia: a guided walk and a free exploration.

3.3 Motion Capture for Immersive Experiences

The paper *Immersive Motorcycle Visualization Using Motion Capture: A Proof of Concept for an Online Motorcycle Configurator*

explores a proof-of-concept prototype that integrates motion capture technology into a VR-based online motorcycle configurator to enhance user experience. Traditional configurators in the motorcycle industry often lack dynamic and immersive interactions, particularly between the rider and the motorcycle. Drawing inspiration from racing games, the prototype incorporates realistic rider-motorcycle interactions, a virtual avatar for user representation, and optimized workflows for adapting motion animations across different models. This approach aims to bridge the gap between static configurators and experiential customization tools, offering a more engaging, personalized, and interactive solution for the motorcycle industry.

4 FACTS & FIGURES

The interdisciplinary ANIVAE workshop series has been an important platform for exploring the intersection of animation, VR, and AR since 2018. The series with its inaugural workshop, co-located with the IEEE VR conference 2018 in Reutlingen, Germany. Since then, ANIVAE has been held annually alongside the IEEE VR conferences, providing a space for researchers, practitioners, and artists to discuss the evolving role of animation in immersive environments.

The workshop has traveled across the globe, with previous editions held in Osaka, Japan (2019), Atlanta, USA (2020, virtually), Lisbon, Portugal (2021, virtually), Christchurch, New Zealand (2022, virtually), Shanghai, China (2023), and Orlando, USA (2024). So far, the workshop series provided a platform for the presentation of 45 papers and for 5 invited speakers, contributing to its success.

The 8th edition of ANIVAE takes place in March 2025 in Saint-Malo, France (2025), co-located with the 2025 edition of the IEEE VR conference, and continues its tradition of exploring innovative uses of animation in AR/VR and pushing the boundaries of interdisciplinary research in this dynamic field. The workshop remains a key event for advancing the understanding of animated content in immersive environments, bridging gaps between technology, creativity, and cultural exploration.

5 CONCLUSION

The interdisciplinary approach of ANIVAE remains central to advance the field of animation in immersive media. By bridging the gap between creative exploration and technical innovation, the workshop facilitates new avenues for both artistic expression and research. The discussions and diverse contributions from the global community of researchers, artists, and technologists continue to push boundaries, drive the development of more immersive, dynamic, and meaningful experiences and lead the way for the future of animation in AR/VR.

ACKNOWLEDGMENTS

The ANIVAE workshop organizers Thomas Moser, Franziska Proksa and Juergen Hagler wish to thank the program committee members and reviewers for the support. Also, the different editions of the ANIVAE workshop series were supported by a set of local co-organizers, namely Pedro Serrazina (Universidade Lusófona de Lisboa), Vladimir Todorovic (University of Western Australia), and Chunning Maggie Guo (Renmin University of China), which the workshop organizers also wish to thank.

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